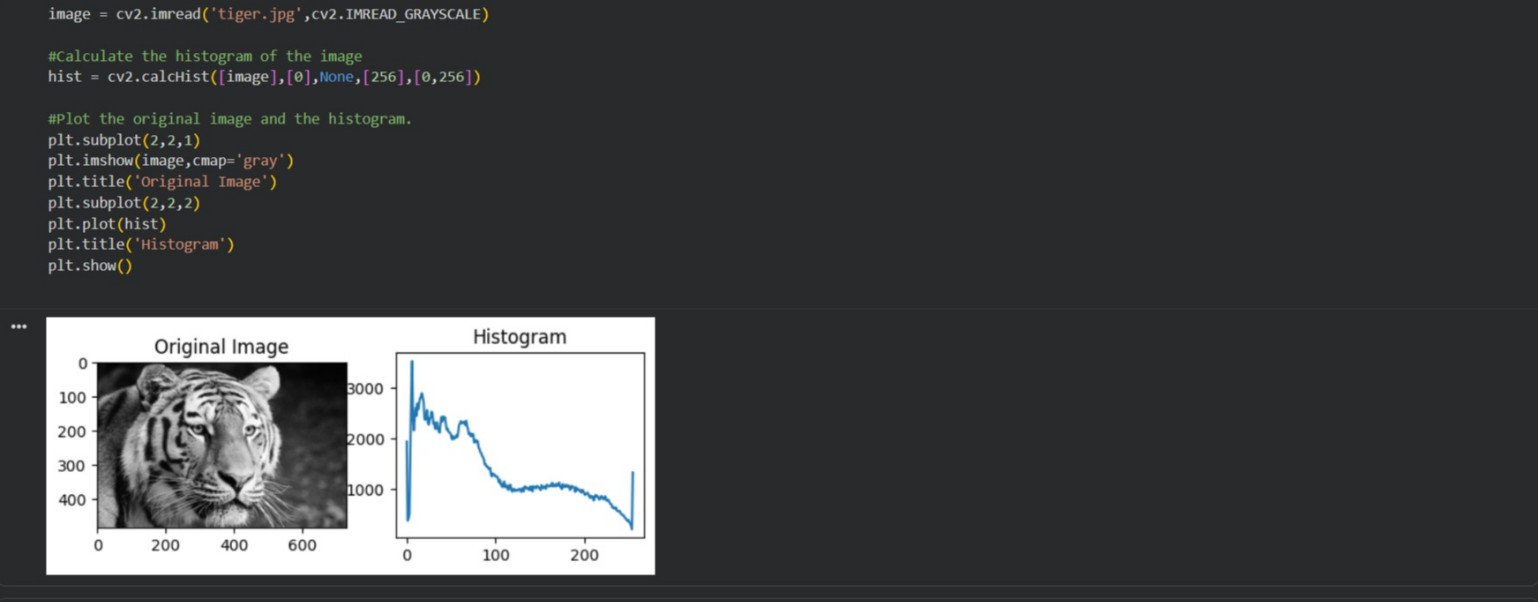


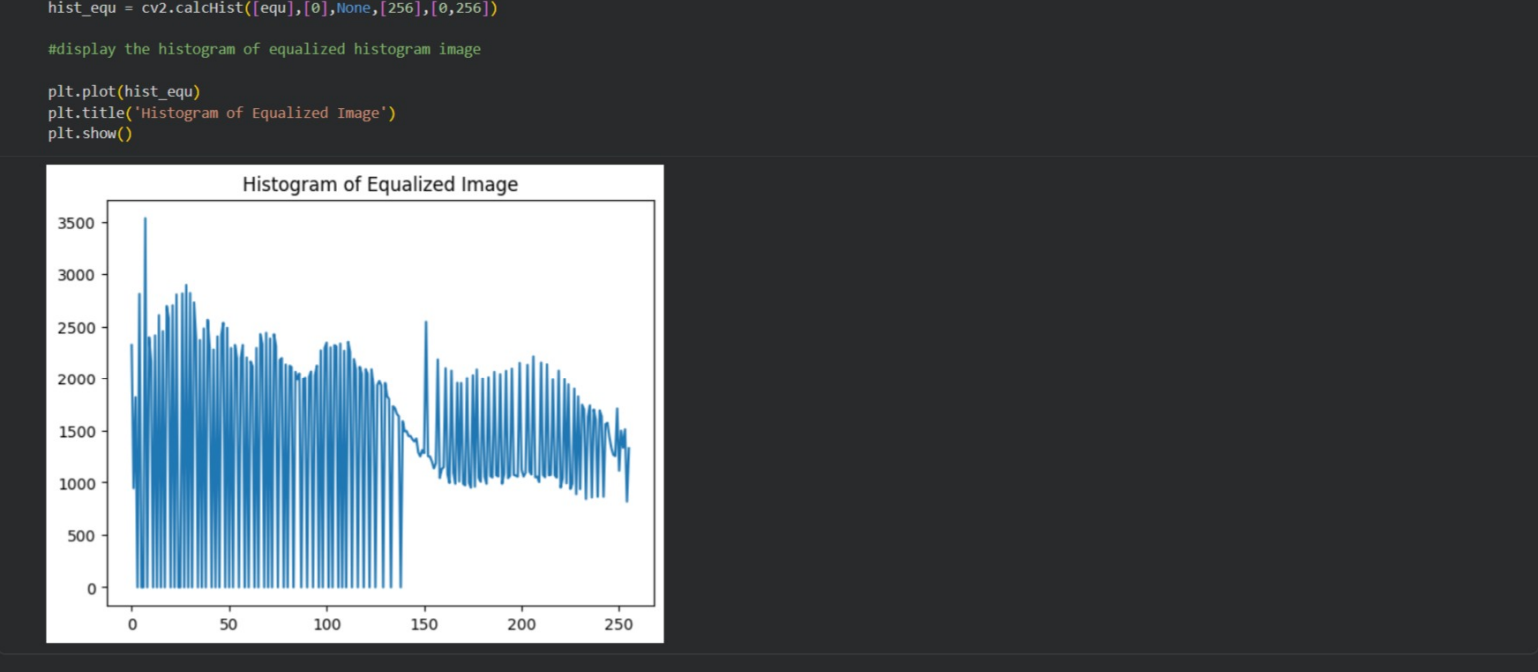
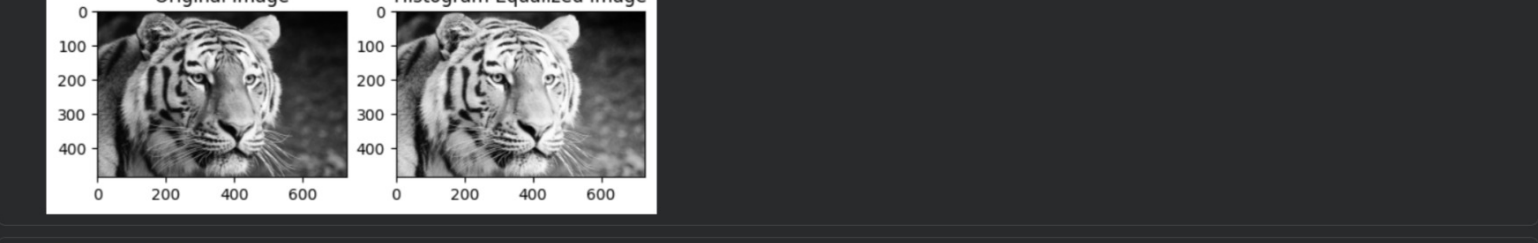
```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab.patches import cv2_imshow
```

Question 01



```
#histogram equalization
equ = cv2.equalizeHist(image)

#display original image and histogram equalized image
plt.subplot(2,2,1)
plt.imshow(image,cmap='gray')
plt.title('Original Image')
plt.subplot(2,2,2)
plt.imshow(equ,cmap='gray')
plt.title('Histogram Equalized Image')
```



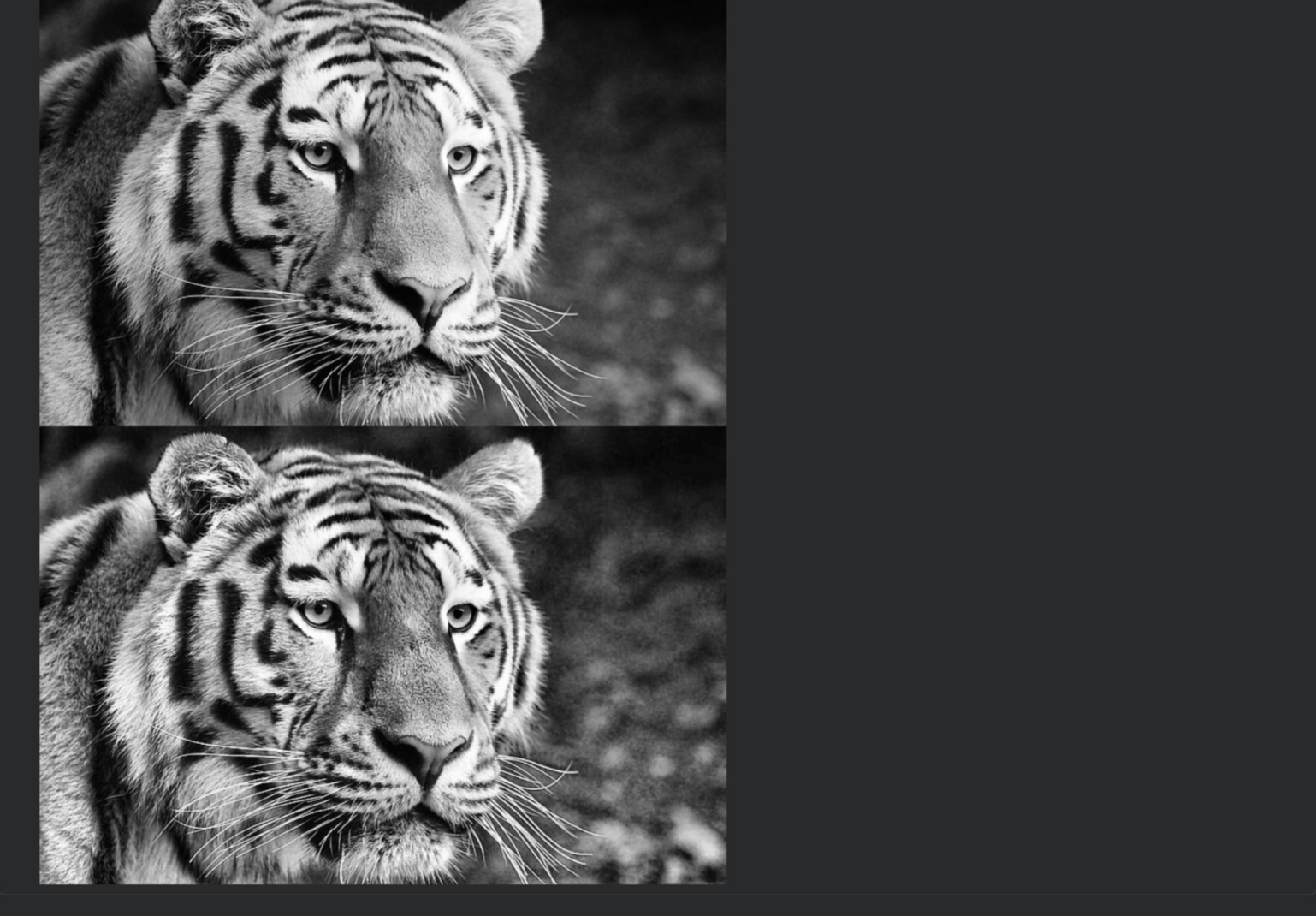
Question 02

```
claheimage = cv2.imread('tiger.jpg',cv2.IMREAD_GRAYSCALE)
cv2_imshow(claheimage)

clahe=cv2.createCLAHE(clipLimit=2.0,tileGridSize=(8,8))

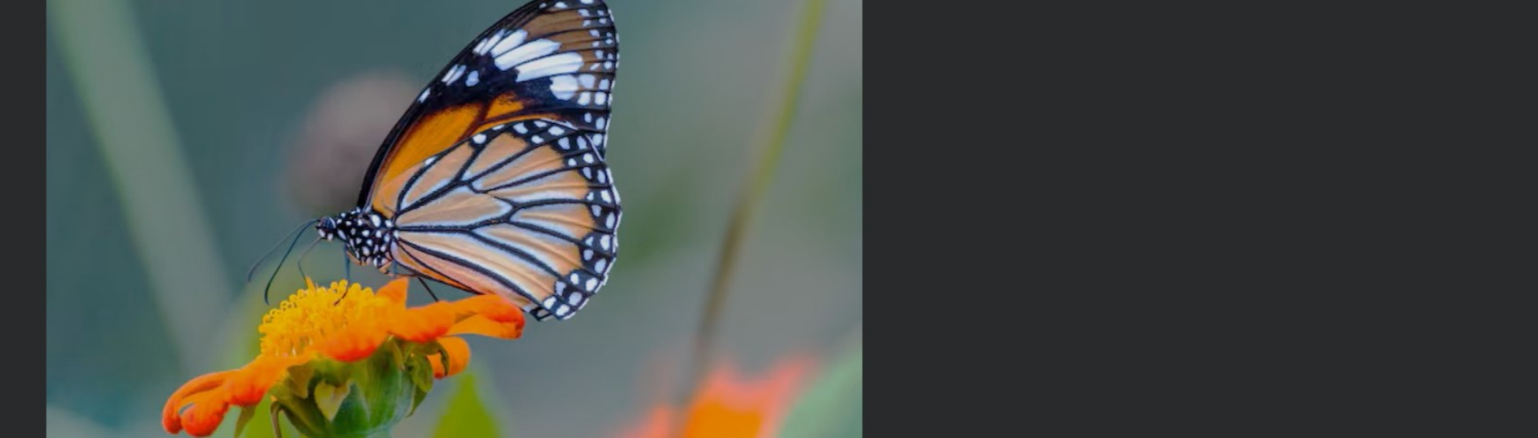
cli=clahe.apply(claheimage)

cv2_imshow(cli)
```



Question 3

```
image_lc= cv2.imread('lcing.jpg',cv2.COLOR_RGB2BGR)
cv2_imshow(image_lc)
```



```
#seperate color channels
B=image_lc[:, :,0] #Blue layer
G=image_lc[:, :,1] #green layer
R=image_lc[:, :,2] #red layer
```

```
#histogram of color channels
B_histo=cv2.calcHist([B],[0],None,[256],[0,256])
G_histo=cv2.calcHist([G],[0],None,[256],[0,256])
R_histo=cv2.calcHist([R],[0],None,[256],[0,256])
```

```
#display histograms
plt.subplot(2,2,1)
plt.plot(B_histo, 'b')

plt.subplot(2,2,2)
plt.plot(G_histo, 'g')

plt.subplot(2,2,3)
plt.plot(R_histo, 'r')

plt.show()
```



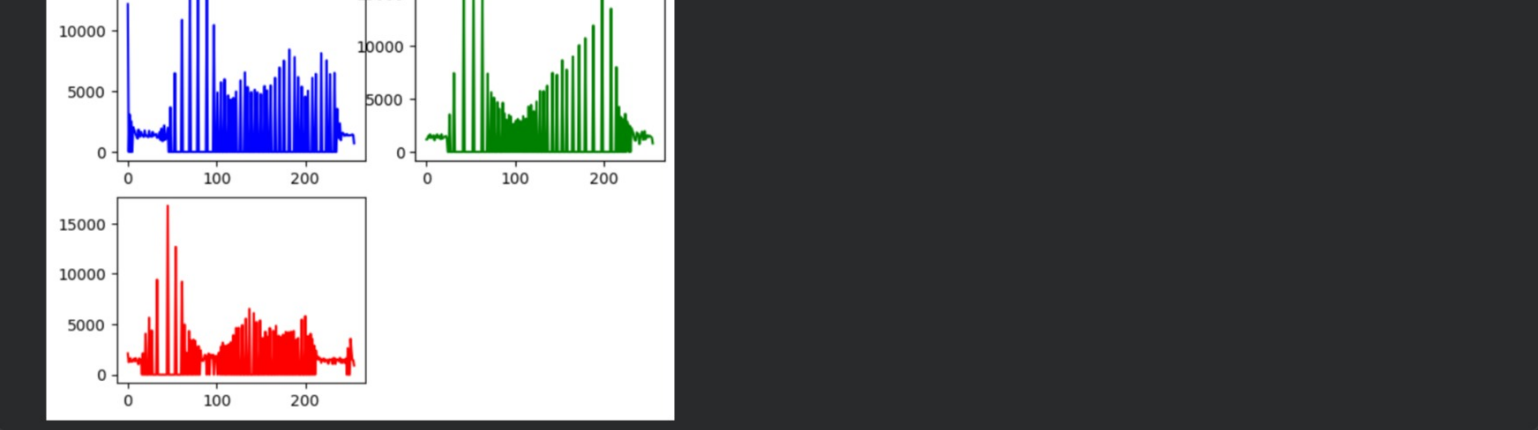
```
#equalize color channels
B_eq=cv2.equalizeHist(B)
G_eq=cv2.equalizeHist(G)
R_eq=cv2.equalizeHist(R)
```

```
#equilized histograms
B_histo_eq=cv2.calcHist([B_eq],[0],None,[256],[0,256])
G_histo_eq=cv2.calcHist([G_eq],[0],None,[256],[0,256])
R_histo_eq=cv2.calcHist([R_eq],[0],None,[256],[0,256])
```

```
#show equalized histograms
plt.subplot(2,2,1)
plt.plot(B_histo_eq, 'b')

plt.subplot(2,2,2)
plt.plot(G_histo_eq, 'g')

plt.subplot(2,2,3)
plt.plot(R_histo_eq, 'r')
```



```
#merge the channels and create new image
image_eq=cv2.merge((B_eq,G_eq,R_eq))
cv2_imshow(image_lc)
cv2_imshow(image_eq)
```

